

Rudransh Joshi

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AI/ML Engineer
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EDUCATION

- Uttarakhand Technical University** Dehradun, India
Bachelor of Technology - Artificial Intelligence & Machine Learning — GPA: 8.59 Sep 2022 - Jul 2026
Courses: Data Structures, Algorithms, Operating Systems, Databases, Digital Logic, Computer Organization & Architecture, Automata Theory, Compiler Design, Networking, Artificial Intelligence, Machine Learning, Data Science, Genetic Algorithms, Deep Learning

SKILLS SUMMARY

- Languages:** Python, TypeScript, C, SQL, Bash
- AI/ML:** PyTorch, TensorFlow, ONNX, LangChain, NumPy, Pandas, Matplotlib, Scikit-Learn, Gymnasium
- Frameworks:** Django, FastAPI, Next.js, React, TailwindCSS, Qt, OpenCV, FFMpeg
- DevOps:** Git, GitHub, CI/CD, GNU/Linux, AWS/GCP/Azure, Nginx, Docker
- Embedded:** Arduino, ESP, Raspberry

EXPERIENCE

- Confidential Client** Remote — Delhi, India
ML Engineer (Freelance) Jan 2026 - Present
 - Built and deployed a computer vision system for automated quality assessment in a large-scale supply chain workflow, training a custom Ultralytics/YOLOv26 segmentation model on ~3K hand-labeled samples **achieving mAP50-95 score of ~0.93** with FP16 quantization to enable **real-time batch-level analysis**.
 - Implemented an edge-first inference pipeline using ONNX Runtime Web with API fallback and developed an analytics dashboard supporting **near-real-time detection under ~400ms** and data-driven QA decisions.
- E Giraffes Digital Pvt. Ltd.** Remote — Pune, India
AI Automation Engineer (Intern) Jun 2024 - Aug 2024
 - Designed standardized AI asset-generation workflows, **reducing onboarding time by ~30%** and minimizing manual intervention across creative pipelines.
 - Automated task coordination using Microsoft Power Automate (SharePoint + Teams), enabling real-time approval workflows and **reducing manual effort by ~50%**.
- Cyber Efficient** Remote — Texas, USA
Full Stack Developer (Part-Time) Dec 2020 - Feb 2022
 - Architected and deployed CyberEfficient.io using Django and Bootstrap, optimizing backend queries to achieve **~35% faster load performance**.
 - Developed a multi-purpose Discord bot **serving 10K+ users with ~20ms avg latency** and a URL shortener built with Flask **handling 25K+ links** with optimized queries for efficient link redirection.

PROJECTS

- StreaX – Offline Productivity PWA (Next.js, React, PWA):** Built a self-hostable, retro-themed productivity tracker with Pomodoro sessions, streak mechanics, and goal gamification, designed for **100% offline on-device data persistence**. Implemented PWA capabilities, installability, and responsive state management using Next.js, React & RetroUI.
- RLSwarm – Physics-Based Swarm RL Environment (Python, Gymnasium, PyGame):** Researched and developed a continuous swarm foraging simulator built on Gymnasium. Benchmarked PPO and DQN agents in scalable multi-agent settings, demonstrating **zero-shot generalization from N → 2N agents** under fixed-area constraints, **retaining ~30% of baseline reward** under doubled agent density.
- while(MOO) – Offline Cattle Breed Identification System (PyTorch, Next.js):** Implemented an offline-first breed classification system using a custom explainable CNN (MooNet) with **~96.7% accuracy** built over MobileNetV4, with feature engineering & preprocessing optimized for edge inference via ONNX Web Runtime. Integrated a PWA dashboard with domain-specific chatbot (AskMoo) and structured breed knowledge base (MOOpedia) using FastAPI and Gemini API.
- how-cli – Context-Aware AI Terminal Assistant (Python, LangChain, Typer):** Developed a cross-platform CLI assistant that converts natural language prompts into OS-specific sequential shell commands using LangChain orchestration. Implemented structured output formatting and context awareness for Linux, macOS, and Windows environments.

PUBLICATIONS

- Joshi et al., Policy Optimization for Scalable Swarm Robotics in Physics-Based Environments:** Under review, 2026. Multi-Agent Systems, Reinforcement Learning, Swarm Intelligence.

HONORS AND AWARDS

- 1st Place, Smart India Hackathon - 2023, 2025
- 2nd Place, UTU' Socio Hackathon - Apr 2025
- Best Overall Hack, Hackathon With Lewis - Oct, 2024
- Most Liked Hack, Codédex Summer Hackathon - Jul 2024
- Academic Excellence Award, School Topper (Secondary: 97%; Senior Secondary: 93%) - 2020, 2022